

IsoTech Ink: Initial Cybersecurity Risk Assessment

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1. Executive Summary:

IsoTech Ink currently faces multiple operational and cybersecurity risks due to rapid technology adoption, insufficient cybersecurity governance practices, and hybrid workforce expansion.

The organization's dependence on third-party vendors, cloud services, remote employees, and digital asset repositories increases the likelihood of unauthorized access, phishing attacks, data exposure, and operational disruption.

This assessment identifies key cybersecurity weaknesses, evaluates organizational risk exposure, and recommends priority areas for security improvements.

2. Assessment Scope:

This assessment focused on the following areas:

- Remote and hybrid workforce security
- Access control management
- Employee security awareness
- Security monitoring visibility
- Vendor and third-party risk
- Endpoint logging and monitoring
- Incident response readiness

3. Key Findings:

Finding 1 - Inconsistent Access Controls

Risk Level: High

Employees and executives reported inconsistent access permissions across systems and repositories.

Risks Identified

- Excessive user privileges
- Increased insider threat exposure
- Weak role separation
- Unauthorized data access

Potential Business Impact

- Enforce least privilege access
- Conduct quarterly access reviews
- Implement Role-Based Access Control (RBAC)
- Require MFA for privileged accounts

Finding 2 - Weak Remote Work Security Procedures

Risk Level: High

Hybrid and remote employees currently operate without standardized security procedures for device usage and remote access.

Risks Identified

- Weak password practices
- Unsecured personal devices
- Unsafe remote access habits
- Increased phishing exposure

Potential Business Impact

- Malware infection
- Unauthorized network access
- Credential compromise

Recommendations

- Implement device compliance standards
- Enforce MFA and Conditional Access
- Improve employee awareness training
- Develop a BYOD security policy

Finding 3 - Limited Employee Security Awareness

Risk Level: High

Employees have not received consistent cybersecurity awareness training related to phishing, password hygiene, or safe remote work practices.

Risks Identified

- Increased phishing success rates
- Social engineering attacks
- Credential theft
- Malware infections

Potential Business Impact

- Financial loss
- Reputational damage
- Business email compromise

Recommendations

- Conduct phishing simulation exercises
- Launch phishing simulation exercises
- Develop employee reporting procedures

Finding 4 - Insufficient Security Monitoring & Logging

Risk Level: Medium-High

Current monitoring capabilities lack centralized visibility into suspicious activity and endpoint events.

Risks Identified

- Limited investigations capability
- Delayed threat detection
- Reduced incident response effectiveness

Monitoring Gaps Identified

- Financial login monitoring
- Brute force detection
- DNS monitoring
- Suspicious PowerShell activity
- Endpoint visibility

Recommendations

- Deploy enhanced Windows logging via Sysmon
- Create alert dashboards and SPL queries
- Implement SIEM monitoring using Splunk
- Standardize log ingestion procedures

Finding 5 - Weak Third-Party Vendor Oversight

Risk Level: Medium-High

Current vendor security evaluation procedures are inconsistent and lack formal security review standards.

Risks Identified

- Third-party data exposure
- Supply chain compromise
- Weak vendor security practices

Potential Business Impact

- Operational disruption
- Compliance exposure
- Data breaches through vendors

Recommendations

- Establish minimum security requirements for vendors
- Deploy vendor risk questionnaires
- Create vendor security review procedures

Finding 6 - Incomplete Incident Response Procedures

Risk Level: Medium

The organization currently lacks standardized incident response documentation and escalation procedures.

Risks Identified

- Delayed response times
- Ineffective containment
- Confusion during security incidents

Recommendations

- Define escalation procedures
- Conduct tabletop exercises
- Develop an Incident Response Plan aligned with NIST 800-61

4. Overall Risk Summary:

<u>Risk Area</u>	<u>Risk Level</u>
Access Control	High
Remote Workforce Security	High
Employee Awareness	High
Monitoring & Logging	Medium-High
Vendor Security	Medium-High
Incident Response	Medium

5. Recommended Priority Actions:

Immediate Priorities (0-30 days)

- Implement MFA
- Deploy enhanced logging
- Begin security awareness training
- Improve SIEM monitoring visibility

Mid-Term Priorities (30-90 days)

- Create RBAC framework
- Develop vendor security program
- Build incident response documentation
- Finalize BYOD policy

Long-Term Priorities (90+ days)

- Conduct tabletop exercises
- Improve detection engineering
- Establish recurring access audits
- Mature governance processes

6. Conclusion:

IsoTech Ink demonstrates significant opportunities to improve cybersecurity governance, operational visibility, and employee security practices.

By strengthening monitoring capabilities, formalizing governance procedures, and improving workforce awareness, the organization can significantly reduce operational and cybersecurity risk exposure while improving long-term security maturity.